Title:

Data Recovery and Autopsy Report

Examiner: Christopher Hernandez

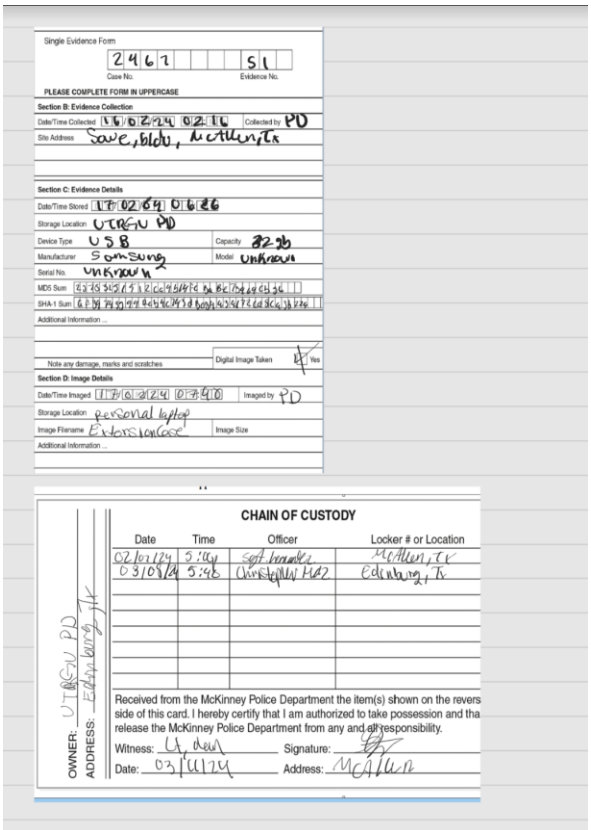
Digital Forensics 4319, Portfolio 4

Executive Summary

In this Report, I have used the Linux OS and Forensic tools such as Scalpel, Foremost, Bulk Extracto and Magic Rescue to successfully recover and investigate files given to me. I will see if the given file has any corrupt or malware in them.

Chain of custody

For the chain of custody, I scanned a paper and pasted it here. Due to the nature of the case, I got it late.



Single Evidence Form

Case No. 2467 Evidence No. S1

PLEASE COMPLETE FORM IN UPPERCASE

Section B: Evidence Collection

Date/Time Collected 16/02/24 02:16 Collected by PD

Site Address Sowe, blvd, McAllen, Tx

Section C: Evidence Details

Date/Time Stored 17/02/24 06:26

Storage Location UTRGV PD

Device Type USB Capacity 32gb

Manufacturer Samsung Model Unknown

Serial No. Unknown

MD5 Sum

SHA-1 Sum

Additional Information ...

Note any damage, marks and scratches

Digital Image Taken ☒ Yes

Section D: Image Details

Date/Time Imaged 17/02/24 07:40 Imaged by PD

Storage Location personal laptop

Image Filename Evidencecase Image Size

Additional Information ...

CHAIN OF CUSTODY

Date	Time	Officer	Locker # or Location
02/02/24	5:40	Sgt Lopez	McAllen, TX
03/03/24	5:46	Christopher Hd2	Edinburg, Tx

Received from the McKinney Police Department the item(s) shown on the revers side of this card. I hereby certify that I am authorized to take possession and tha release the McKinney Police Department from any and all responsibility.

Witness: [illegible] Signature: [illegible]

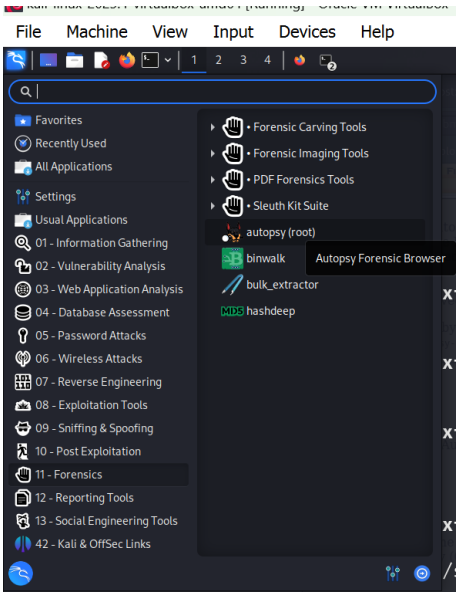
Date: 03/11/24 Address: McAllen

OWNER: UTRGV PD

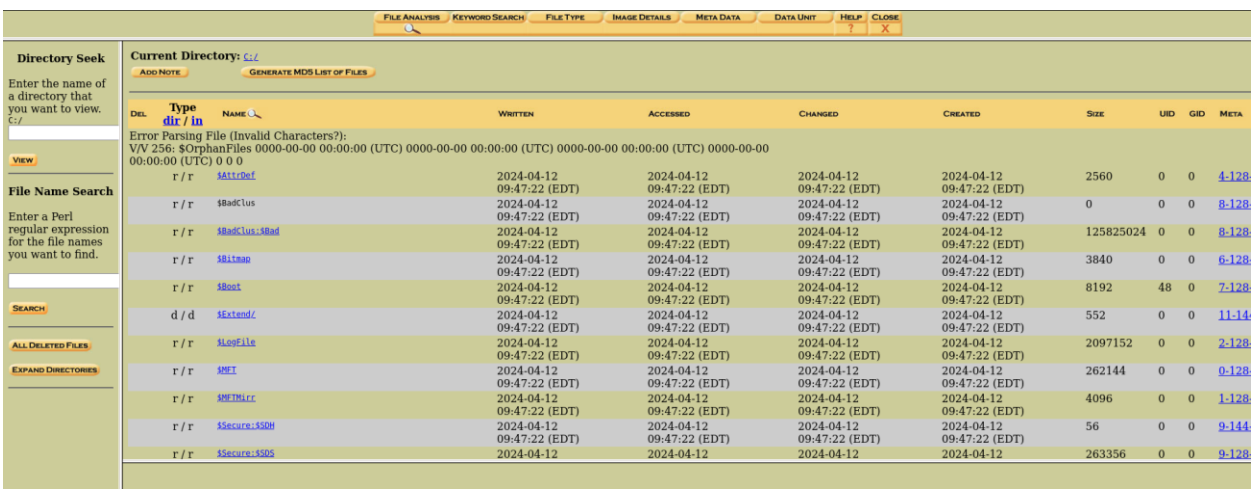
ADDRESS: Edinburg, Tx

Autopsy Analysis

For the Autopsy analysis, I used the autopsy tool that's under forensics tools in Linux.



Then, I created a new case and put all the information necessary for the creation of the image file. After that, this was shown.



I grabbed a random file and checked for the magic number in hex.

PREVI

EXPORT CONTE

ASCII (display - report) * Hex (display

File Type: JPEG image data, JFIF standard 1.01, resolution (DPI), density

Cluster: 9683
Status: Allocated
Find Meta Data Address

Hex Contents of Cluster 9683 in Autopsy-case-24.001-0-0

```
0       ffd8ffe0 00104a46 49460001 01010078     .... ..JF IF.. ...x
16      00780000 ffdb0043 000a0707 0807060a     .x.. ...C .... ....
32      0808080b 0a0a0b0e 18100e0d 0d0e1d15     .... .... .... ....
48      16111823 1f252422 1f222126 2b372f26     ...# .%$" ."!& +7/&
64      29342921 22304131 34393b3e 3e3e252e     )4)! "0A1 49;> >>%.
80      4449433c 48373d3e 3bffdb00 43010a0b     DIC< H7=> ;... C...
96      0b0e0d0e 1c10101c 3b282228 3b3b3b3b     .... .... ;("( ;;;;
112     3b3b3b3b 3b3b3b3b 3b3b3b3b 3b3b3b3b     ;;;; ;;;; ;;;; ;;;;
128     3b3b3b3b 3b3b3b3b 3b3b3b3b 3b3b3b3b     ;;;; ;;;; ;;;; ;;;;
144     3b3b3b3b 3b3b3b3b 3b3b3b3b 3b3bffc0     ;;;; ;;;; ;;;; ;;..
160     00110801 8a030c03 01220002 11010311     .... .... ."".. ....
176     01ffc400 1f000001 05010101 01010100     .... .... .... ....
192     00000000 00000001 02030405 06070809     .... .... .... ....
208     0a0bffc4 00b51000 02010303 02040305     .... .... .... ....
224     05040400 00017d01 02030004 11051221     .... ..}. .... ...!
240     31410613 51610722 71143281 91a10823     1A.. Qa." q.2. ...#
256     42b1c115 52d1f024 33627282 090a1617     B... R..$ 3br. ....
272     18191a25 26272829 2a343536 3738393a     ...% &'() *456 789:
288     43444546 4748494a 53545556 5758595a     CDEF GHIJ STUV WXYZ
304     63646566 6768696a 73747576 7778797a     cdef ghij stuv wxyz
320     83848586 8788898a 92939495 96979899     .... .... .... ....
336     9aa2a3a4 a5a6a7a8 a9aab2b3 b4b5b6b7     .... .... .... ....
352     b8b9bac2 c3c4c5c6 c7c8c9ca d2d3d4d5     .... .... .... ....
368     d6d7d8d9 dae1e2e3 e4e5e6e7 e8e9eaf1     .... .... .... ....
384     f2f3f4f5 f6f7f8f9 faffc400 1f010003     .... .... .... ....
400     01010101 01010101 01000000 00000001     .... .... .... ....
416     02030405 06070809 0a0bffc4 00b51100     .... .... .... ....
432     02010204 04030407 05040400 01027700     .... .... .... ..w.
448     01020311 04052131 06124151 07617113     .... ..!1 ..AQ .aq.
464     22328108 144291a1 b1c10923 3352f015     "2.. .B.. ...# 3R..
480     6272d10a 162434e1 25f11718 191a2627     br.. .$4. %... ..&'
496     28292a35 36373839 3a434445 46474849     ()*5 6789 :CDE FGHI
512     4a535455 56575859 5a636465 66676869     JSTU VWXY Zcde fghi
528     6a737475 76777879 7a828384 85868788     istu vwxv z... ....
```

\

This image is the general details of the file. It gives info such as the metadata, file system type, sector size, clusters, and root directory.

General File System Details

FILE SYSTEM INFORMATION

File System Type: NTFS
Volume Serial Number: CC48F26C48F254A4
OEM Name: NTFS
Volume Name: Intellectual Property
Version: Windows XP

METADATA INFORMATION

First Cluster of MFT: 10240
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 256
Root Directory: 5

CONTENT INFORMATION

Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 30718
Total Sector Range: 0 - 245758

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-578 Flags: Resident,Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:

VirusTotal

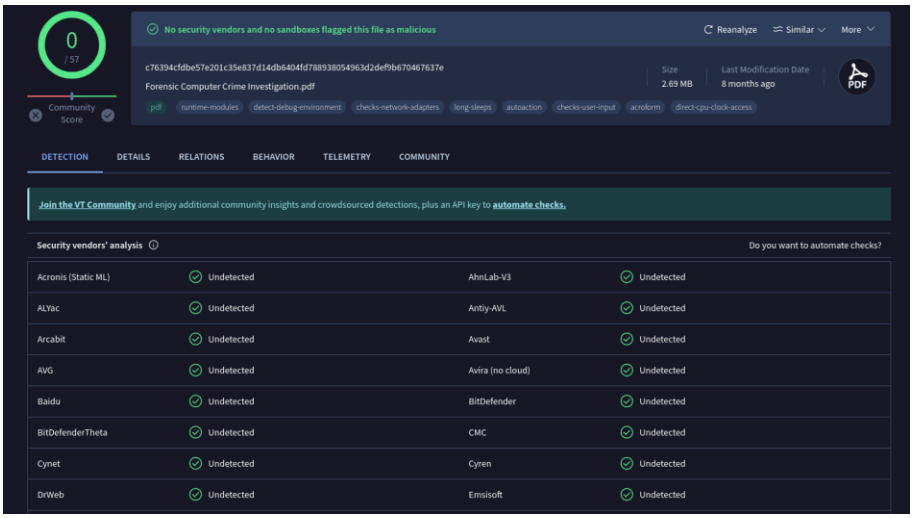
The following images are the autopsy I did on the given file. For this image, I generated the md5 list of the files.

```
MD5 Values for files in C:/ (Autopsy-case-24.001-0-0)

ad617ac3906958de35eacc3d90d31043  -     $AttrDef
d41d8cd98f00b204e9800998ecf8427e  -     $BadClus
d41d8cd98f00b204e9800998ecf8427e  -     $BadClus:$Bad
38f5cc838b0bf38466dba2594e1449a8  -     $Bitmap
587394ecdbb4c315649161f293243a9c  -     $Boot
8632d2342f11a005276dcc233d475834  -     $LogFile
c67cba004a111f53850febf03000922e  -     $MFT
3272308bc6f3b7330c9e914347c00f89  -     $MFTMirr
3ae634c9b81c24b96a75706602726568  -     $Secure:$SDS
29c8d340eedb44039c942149ee9fea72  -     $Secure:$SDH
0ef04368ef411190e098df2d950ff15a  -     $Secure:$SII
7ff498a44e45e77374cc7c962b1b92f2  -     $UpCase
cbe523418d3183f2099a1ee8490c08d5  -     $UpCase:$Info
d41d8cd98f00b204e9800998ecf8427e  -     $Volume
379d4f73c446125abd806b25e5992e62  -     180465960-7522424-Incident-Response-Computer-Forensics-Toolk.pdf
b99f4757239c79f918c50b6ff3015d3b  -     49189758-forensics.pdf
98c52b25774178f025958af966c9106a  -     94272864-Forensic-Computer-Crime-Investigation.pdf
8f2b91124250ed00cdc431e7de2ea310  -     CMakeLists.txt
4f9e75a41d02666cd5cc86bd33a578fe  -     HxDSetup.exe
e744a06089db576595165bc5910b655c  -     image005.jpg
76a3361168d4565db7a856c144edbee0  -     image006.png
8bd950341868e7ef5760ee5da93802d7  -     image007.png
f90f411aca3436e5b754704461e03686  -     image008.jpg
e2cadfdbacd3b4f1afb7f275ce09c29c  -     image009.png
18871f97e1154c5bd94e70e3b4a16086  -     image010.png
```

I went ahead and tested the file for any malware for a general conclusion on whether these are overall healthy files or contain malware.

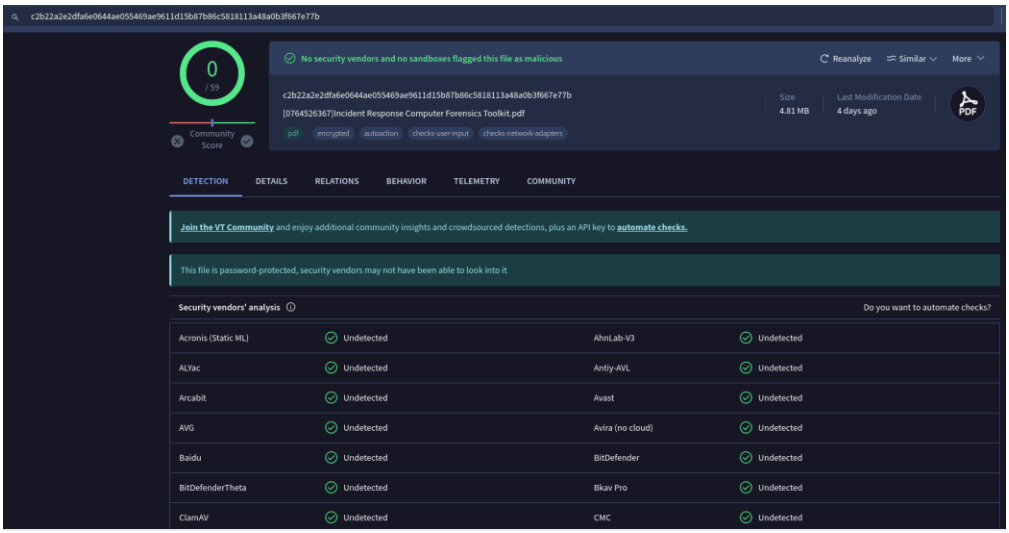

```
MD5 Values for files in C:/ (Autopsy-case-24.001-0-0)

ad617ac3906958de35eacc3d90d31043  -     $AttrDef
d41d8cd98f00b204e9800998ecf8427e  -     $BadClus
d41d8cd98f00b204e9800998ecf8427e  -     $BadClus:$Bad
38f5cc838b0bf38466dba2594e1449a8  -     $Bitmap
587394ecdbb4c315649161f293243a9c  -     $Boot
8632d2342f11a005276dcc233d475834  -     $LogFile
c67cba004a111f53850febf03000922e  -     $MFT
3272308bc6f3b7330c9e914347c00f89  -     $MFTMirr
3ae634c9b81c24b96a75706602726568  -     $Secure:$SDS
29c8d340eedb44039c942149ee9fea72  -     $Secure:$SDH
0ef04368ef411190e098df2d950ff15a  -     $Secure:$SII
7ff498a44e45e77374cc7c962b1b92f2  -     $UpCase
cbe523418d3183f2099a1ee8490c08d5  -     $UpCase:$Info
d41d8cd98f00b204e9800998ecf8427e  -     $Volume
379d4f73c446125abd806b25e5992e62  -     180465960-7522424-Incident-Response-Computer-Forensics-Toolk.pdf
b99f4757239c79f918c50b6ff3015d3b  -     49189758-forensics.pdf
98c52b25774178f025958af966c9106a  -     94272864-Forensic-Computer-Crime-Investigation.pdf
8f2b91124250ed00cdc431e7de2ea310  -     CMakeLists.txt
4f9e75a41d02666cd5cc86bd33a578fe  -     HxDSetup.exe
e744a06089db576595165bc5910b655c  -     image005.jpg
76a3361168d4565db7a856c144edbee0  -     image006.png
8bd950341868e7ef5760ee5da93802d7  -     image007.png
f90f411aca3436e5b754704461e03686  -     image008.jpg
e2cadfdbacd3b4f1afb7f275ce09c29c  -     image009.png
18871f97e1154c5bd94e70e3b4a16086  -     image010.png
```


VirusTotal did not find anything malicious.



I did the same thing with another file. And it was a healty file.

```
MD5 Values for files in C:/ (Autopsy-case-24.001-0-0)

ad617ac3906958de35eacc3d90d31043  -     $AttrDef
d41d8cd98f00b204e9800998ecf8427e  -     $BadClus
d41d8cd98f00b204e9800998ecf8427e  -     $BadClus:$Bad
38f5cc838b0bf38466dba2594e1449a8  -     $Bitmap
587394ecdbb4c315649161f293243a9c  -     $Boot
8632d2342f11a005276dcc233d475834  -     $LogFile
c67cba004a111f53850febf03000922e  -     $MFT
3272308bc6f3b7330c9e914347c00f89  -     $MFTMirr
3ae634c9b81c24b96a75706602726568  -     $Secure:$SDS
29c8d340eedb44039c942149ee9fea72  -     $Secure:$SDH
0ef04368ef411190e098df2d950ff15a  -     $Secure:$SII
7ff498a44e45e77374cc7c962b1b92f2  -     $UpCase
cbe523418d3183f2099a1ee8490c08d5  -     $UpCase:$Info
d41d8cd98f00b204e9800998ecf8427e  -     $Volume
379d4f73c446125abd806b25e5992e62  -     180465960-7522424-Incident-Response-Computer-Forensics-Toolk.pdf
b99f4757239c79f918c50b6ff3015d3b  -     49189758-forensics.pdf
98c52b25774178f025958af966c9106a  -     94272864-Forensic-Computer-Crime-Investigation.pdf
8f2b91124250ed00cdc431e7de2ea310  -     CMakeLists.txt
4f9e75a41d02666cd5cc86bd33a578fe  -     HxDSetup.exe
e744a06089db576595165bc5910b655c  -     image005.jpg
76a3361168d4565db7a856c144edbee0  -     image006.png
8bd950341868e7ef5760ee5da93802d7  -     image007.png
f90f411aca3436e5b754704461e03686  -     image008.jpg
e2cadfdbacd3b4f1afb7f275ce09c29c  -     image009.png
18871f97e1154c5bd94e70e3b4a16086  -     image010.png
```



For the next image, I got the report for a random file to see if everything was shown correctly.

```
                  Autopsy ascii Cluster Report

-------------------------------------------------------------------
                    GENERAL INFORMATION

Cluster: 14
Cluster Size: 4096

Pointed to by MFT Entry: 10-128-1
Pointed to by files:
  C://$UpCase
MD5 of raw Cluster: b34a71c0acad32a05c2381fed7f9ecd9  -
MD5 of ascii output: c2efaae8185982e4510fd6327d983443  -

Image: '/var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001'
Offset: Full image
File System Type: ntfs

Date Generated: Sun Apr 21 16:47:48 2024
Investigator: christopher
-------------------------------------------------------------------
                        CONTENT

.X.X.X.X.X.X.X.X.X      X
X.X.X
X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X X!X"X#X$X%X&X'X(X)X*X+X,X-
X.X/X0X1X2X3X4X5X6X7X8X9X:X;X<X=X>X?X@XAXBXCXDXEXFXGXHXIXJXKXLXMXNXOXPXQXRXSXTXUXVXWXXXYXZX[X\X
.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X.X
Y.Y.Y
```

Sorter Output

The sorter output gave me this info.

---

sorter output

Images

- /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001

Files (57)

Files Skipped (13)

- Non-Files (13)
- Reallocated Name Files (0)
- 'ignore' category (0)

Extensions

- Extension Mismatches (8)

Categories (44)

- archive (0)
- audio (0)
- compress (0)
- crypto (0)
- data (17)
- disk (1)
- documents (4)
- exec (1)
- images (10)
- system (0)
- text (9)
- unknown (2)
- video (0)

Here are the extracted images. And some documents

C:/$Extend/$RmMetadata/$TxfLog/$TxfLog.blf
Targa image data - Map 33355 x 50764 x 1 ""
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 33-128-1

C:/$Extend/$RmMetadata/$TxfLog/$TxfLogContainer00000000000000000001
Targa image data - Map 65536 x 65536 x 1 ""
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 34-128-1

C:/$UpCase
Targa image data - Map 6 x 7 x 8 +4 +5
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 10-128-1

C:/bad files/calibre_cover.jpg
JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length 16, baseline, precision 8, 717x900, components 3
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 47-128-1

C:/image005.jpg
JPEG image data, JFIF standard 1.01, resolution (DPI), density 120x120, segment length 16, baseline, precision 8, 780x394, components 3
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 49-128-1

C:/image006.png
PNG image data, 91 x 67, 8-bit/color RGBA, non-interlaced
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 52-128-3

C:/image007.png
PNG image data, 1920 x 1080, 8-bit/color RGB, non-interlaced
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 53-128-1

C:/image008.jpg
JPEG image data, JFIF standard 1.01, resolution (DPI), density 120x120, segment length 16, baseline, precision 8, 780x439, components 3
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 40-128-1

C:/image009.png
PNG image data, 377 x 67, 8-bit/color RGBA, non-interlaced
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 41-128-3

C:/image010.png
PNG image data, 1920 x 1080, 8-bit/color RGB, non-interlaced
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 48-128-1

documents Category

C:/180465960-7522424-Incident-Response-Computer-Forensics-Toolk.pdf
PDF document, version 1.4, 362 page(s)
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 44-128-1

C:/49189758-forensics.pdf
PDF document, version 1.5, 243 page(s)
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 42-128-1

C:/94272864-Forensic-Computer-Crime-Investigation.pdf
PDF document, version 1.4, 24 page(s)
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 43-128-1

C:/bad files/Windows Registry Forensics.pdf
PDF document, version 1.6, 225 page(s)
Image: /var/lib/autopsy/Autopsy-case-24.001/host1/images/Autopsy-case-24.001 Inode: 50-128-1

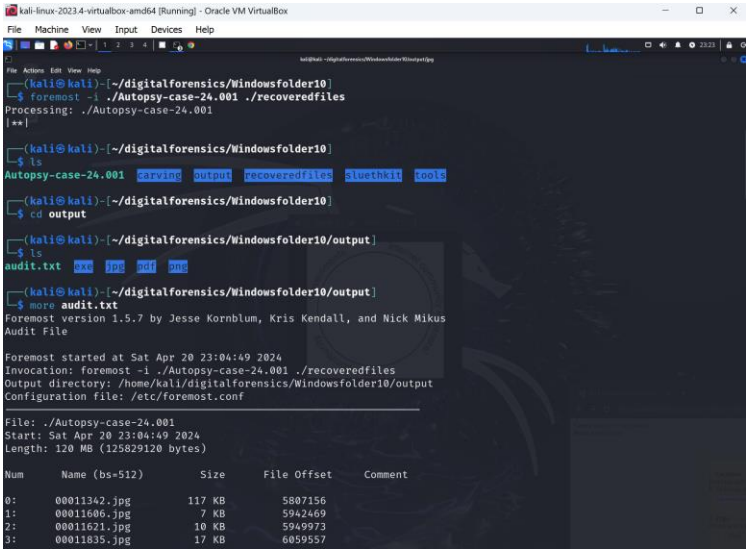
Foremost Tool

For this tool, I used the shown commands to extract files. They were all sent to the output directory, and this was the result of the tool usage.

As seen, I extracted 48 jpg files, 1exe file, 4 png files and 7 pdf files.

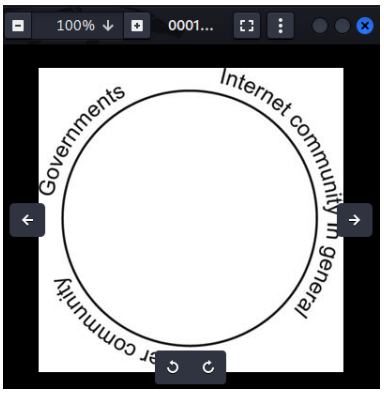
```
File  Actions  Edit  View  Help
34:     00064465.jpg     19 KB     33006245
35:     00064548.jpg     20 KB     33048586
36:     00064588.jpg      7 KB     33069334
37:     00065579.jpg     38 KB     33576454
38:     00065874.jpg     16 KB     33727813
39:     00065965.jpg     18 KB     33774398
40:     00066195.jpg     23 KB     33891902
41:     00066290.jpg      8 KB     33940641
42:     00066455.jpg     31 KB     34025122
43:     00066645.jpg     30 KB     34122540
44:     00066835.jpg     22 KB     34219534
45:     00066896.jpg      9 KB     34251248
46:     00077368.jpg     40 KB     39612416
47:     00077464.jpg     34 KB     39661568
48:     00018064.exe    118 KB      9248768    06/14/2018 13:27:46
49:     00015984.png    582 KB      8183808     (1920 x 1080)
50:     00057040.png    583 KB     29204480     (1920 x 1080)
51:     00077456.png      2 KB     39657472     (377 x 67)
52:     00077536.png      1 KB     39698432     (91 x 67)
53:     00011328.pdf      2 MB      5799936     (PDF is Linearized)
54:     00024800.pdf      5 MB     12697600
55:     00036256.pdf      2 MB     18563072     (PDF is Linearized)
56:     00041672.pdf      2 MB     21336064     (PDF is Linearized)
57:     00047184.pdf      2 MB     24158208
58:     00060696.pdf      3 MB     31076352
59:     00112640.pdf     18 MB     57671680     (PDF is Linearized)
Finish: Sat Apr 20 23:04:56 2024

60 FILES EXTRACTED

jpg:= 48
exe:= 1
png:= 4
pdf:= 7
```

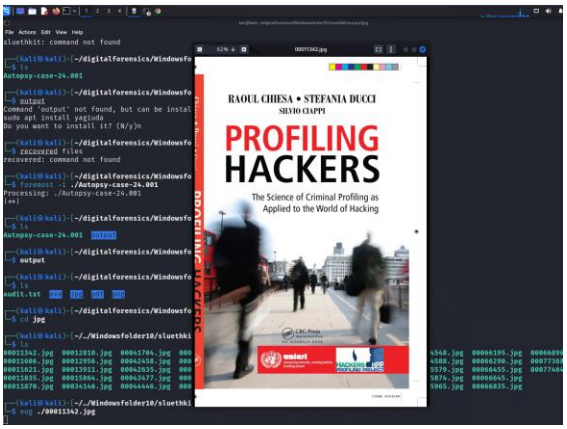
```
File  Actions  Edit  View  Help
Foremost version 1.5.7 by Jesse Kornblum, Kris Kendall, and Nick Mikus
Audit File

Foremost started at Sat Apr 20 23:04:49 2024
Invocation: foremost -i ./Autopsy-case-24.001 ./recoveredfiles
Output directory: /home/kali/digitalforensics/Windowsfolder10/output
Configuration file: /etc/foremost.conf
*******************************************************************
File: ./Autopsy-case-24.001
Start: Sat Apr 20 23:04:49 2024
Length: 120 MB (125829120 bytes)

Num      Name (bs=512)         Size      File Offset     Comment

0:      00011342.jpg        117 KB        5807156
1:      00011606.jpg          7 KB        5942469
2:      00011621.jpg         10 KB        5949973
3:      00011835.jpg         17 KB        6059557
4:      00011870.jpg         11 KB        6077853
5:      00012810.jpg         10 KB        6558846
6:      00012956.jpg          7 KB        6633628
7:      00013911.jpg        176 KB        7122695
8:      00015864.jpg         59 KB        8122368
9:      00034146.jpg         35 KB       17482945
10:     00041704.jpg        108 KB       21352537
11:     00042458.jpg         76 KB       21738563
12:     00042635.jpg         44 KB       21829151
13:     00043477.jpg         38 KB       22260383
14:     00044446.jpg          6 KB       22756482
15:     00044464.jpg         12 KB       22765681
16:     00044521.jpg          4 KB       22794821
17:     00045567.jpg          9 KB       23330569
18:     00045593.jpg         15 KB       23343932
19:     00045754.jpg         27 KB       23426415
20:     00045813.jpg         26 KB       23456633
21:     00061782.jpg         37 KB       31632655
```

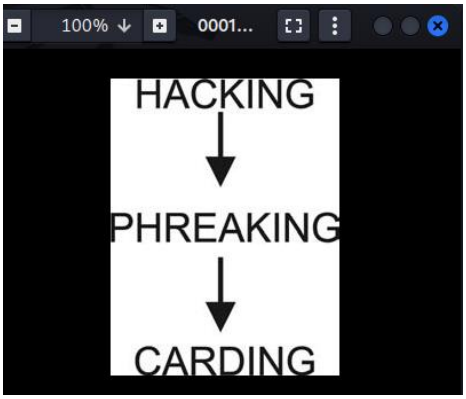


Extracted images:

Some images of the extraction are:

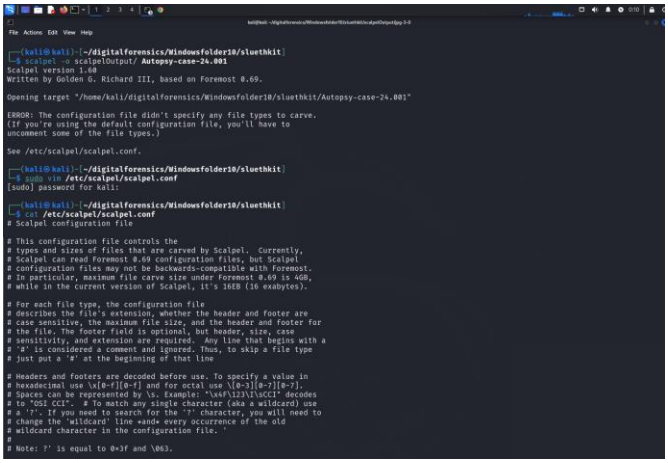






Scalpel Tool

In these pictures, I used the required commands for the tool and uncommented after using the vim command to make the extracted file show. This tool is very similar to the last one.



I uncommented the gif and jpg.

```
#
#-----------------------------------------------------------------
# GRAPHICS FILES
#-----------------------------------------------------------------
#
#
# AOL ART files
#       art     y       150000  \x4a\x47\x04\x0e        \xcf\xc7\xcb
#       art     y       150000  \x4a\x47\x03\x0e        \xd0\xcb\x00\x00
#
# GIF and JPG files (very common)
        gif     y       5000000         \x47\x49\x46\x38\x37\x61        \x00\x3b
        gif     y       5000000         \x47\x49\x46\x38\x39\x61        \x00\x3b
        jpg     y       5242880         \xff\xd8\xff???Exif             \xff\xd9        REVERSE
        jpg     y       5242880         \xff\xd8\xff???JFIF             \xff\xd9        REVERSE
#
#
# PNG
#       png     y       20000000        \x50\x4e\x47?   \xff\xfc\xfd\xfe
#
#
# BMP   (used by MSWindows, use only if you have reason to think there are
#       BMP files worth digging for. This often kicks back a lot of false
#       positives
```

Here is the total of images I extracted:

```
┌──(kali㉿kali)-[~/digitalforensics/Windowsfolder10/sluethkit/scalpelOutput]
└─$ ls
udit.txt  jpg-3-0

┌──(kali㉿kali)-[~/digitalforensics/Windowsfolder10/sluethkit/scalpelOutput]
└─$ cd jpg-3-0

┌──(kali㉿kali)-[~/…/Windowsfolder10/sluethkit/scalpelOutput/jpg-3-0]
└─$ ls
00000000.jpg  00000005.jpg  00000010.jpg  00000015.jpg  00000020.jpg  00000025.jpg  00000030.jpg  00000035.jpg  00000040.jpg  00000045.jpg
00000001.jpg  00000006.jpg  00000011.jpg  00000016.jpg  00000021.jpg  00000026.jpg  00000031.jpg  00000036.jpg  00000041.jpg  00000046.jpg
00000002.jpg  00000007.jpg  00000012.jpg  00000017.jpg  00000022.jpg  00000027.jpg  00000032.jpg  00000037.jpg  00000042.jpg  00000047.jpg
00000003.jpg  00000008.jpg  00000013.jpg  00000018.jpg  00000023.jpg  00000028.jpg  00000033.jpg  00000038.jpg  00000043.jpg
00000004.jpg  00000009.jpg  00000014.jpg  00000019.jpg  00000024.jpg  00000029.jpg  00000034.jpg  00000039.jpg  00000044.jpg

┌──(kali㉿kali)-[~/…/Windowsfolder10/sluethkit/scalpelOutput/jpg-3-0]
└─$
```

Bulk Extracto Tool

In this tool, I used bulk extracto to extract information such as search, text, instagram logs, telephone numbers and passwords.

```
┌──(kali㉿kali)-[~/digitalforensics/Windowsfolder10/sluethkit]
└─$ bulk_extractor -o bulk_output Autopsy-case-24.001

mkdir "bulk_output"
bulk_extractor version: 2.1.0
Input file: "Autopsy-case-24.001"
Output directory: "bulk_output"
Disk Size: 125829120
Scanners: aes base64 elf evtx exif facebook find gzip httplogs json kml_carved msxml net ntfsindx ntfslogfile ntfsmft ntfsusn pdf rar sqlite utmp
vcard_carved windirs winlnk winpe winprefetch zip accts email gps
Threads: 2
going multi-threaded ... ( 2 )
bulk_extractor      Sun Apr 21 00:21:16 2024

available_memory: 756359168
bytes_queued: 0
depth0_bytes_queued: 0
depth0_sbufs_queued: 0
elapsed_time:  0:00:00
```

```
┌──(kali㉿kali)-[~/digitalforensics/Windowsfolder10/sluethkit]
└─$ cd bilk_output
cd: no such file or directory: bilk_output

┌──(kali㉿kali)-[~/digitalforensics/Windowsfolder10/sluethkit]
└─$ cd bulk_output

┌──(kali㉿kali)-[~/digitalforensics/Windowsfolder10/sluethkit/bulk_output]
└─$ ls
aes_keys.txt                email.txt              ip_histogram.txt       ntfsusn_carved.txt      telephone.txt             windirs.txt
alerts.txt                  ether_histogram_1.txt  ip.txt                 pii_teamviewer.txt      unrar_carved.txt          winlnk.txt
ccn_histogram.txt           ether_histogram.txt    jpeg_carved.txt        pii.txt                 url_facebook-address.txt  winpe_carved
ccn_track2_histogram.txt    ether.txt              json.txt               rar.txt                 url_facebook-id.txt       winpe_carved.txt
ccn_track2.txt              evtx_carved.txt        kml_carved.txt         report.xml              url_histogram.txt         winpe.txt
ccn.txt                     exif.txt               ntfsindx_carved        rfc822.txt              url_microsoft-live.txt    winprefetch.txt
domain_histogram.txt        facebook.txt           ntfsindx_carved.txt    sin.txt                 url_searches.txt          zip.txt
domain.txt                  find_histogram.txt     ntfslogfile_carved     sqlite_carved.txt       url_services.txt
elf.txt                     find.txt               ntfslogfile_carved.txt tcp_histogram.txt       url.txt
email_domain_histogram.txt  gps.txt                ntfsmft_carved         tcp.txt                 utmp_carved.txt
email_histogram.txt         httplogs.txt           ntfsmft_carved.txt     telephone_histogram.txt vcard.txt
```

Here are some of the telephone numbers

```
# Filename: Autopsy-case-24.001
# Histogram-File-Version: 1.1
n=20    7036328441
n=10    7036328449
n=6     7036328442
n=4     7036327572
n=4     7036328194
n=4     7039851388
n=4     7039851393
n=3     2023233300
n=3     7036327582
n=3     7036327626
n=3     8002322538
n=2     2023233205
n=2     7036327288
n=2     7036327334
n=2     7036327356
n=2     7036327586
n=2     7036328374
n=2     7036328446
n=2     8885859078
n=2     9787508400
n=1     2025141026
n=1     2025146809
n=1     2522277013
n=1     3046252360
n=1     7036327314
n=1     7036327315
n=1     7036327641
n=1     7036327766
n=1     7036328334
n=1     7036328360
n=1     7036328443
n=1     7036328444
n=1     7039851302
n=1     7039852400
n=1     8007296686
n=1     8007622974
```

```
# Feature-File-Version: 1.1
5963385-PDF-2174        978-750-8400    vers, MA 01923, 978-750-8400. CCC is a not-f
14413531-PDF-2529       800-729-6686     Information at 800-729-6686. 164. Wish ED,
19600270-PDF-630        703-632-8334    e Control Unit, 703-632-8334, prior to submi
19600270-PDF-766        703-632-8360    be directed to  703-632-8360. Requests for e
19623305-PDF-216        703-632-8441     be directed to 703-632-8441. Follow the evi
19625232-PDF-238        703-632-8441     be directed to 703-632-8441. Follow the evi
19627300-PDF-381        703-632-8449     be directed to 703-632-8449. Follow the evi
19627300-PDF-840        703-632-8449    e Laboratory at 703-632-8449 for additional
19629525-PDF-605        703-632-7641     be directed to 703-632-7641. Follow the evi
19633451-PDF-442        703-985-1393     be directed to 703-985-1393. Questions conc
19633451-PDF-514        703-985-1388     be directed to 703-985-1388. Audio examinat
19637310-PDF-329        703-632-8441     be directed to 703-632-8441. Follow the evi
19639281-PDF-243        703-632-8449     be directed to 703-632-8449. Follow the evi
19641266-PDF-500        703-632-8441     be directed to 703-632-8441. Follow the evi
19645466-PDF-209        703-632-8441    e Laboratory at 703-632-8441 prior to submit
19645466-PDF-507        703-632-8441     be directed to 703-632-8441. Follow the evi
19650862-PDF-185        703-985-1302     be directed to 703-985-1302. Follow the evi
19654760-PDF-284        703-632-8441    e Laboratory at 703-632-8441 prior to submit
19654760-PDF-580        703-632-8441     be directed to 703-632-8441. Follow the evi
19656631-PDF-484        703-632-8194     be directed to 703-632-8194.Cryptanalysis a
19656631-PDF-843        703-632-7356    cryption. Call  703-632-7356 or 703-632-7334
19656631-PDF-859        703-632-7334    703-632-7356 or 703-632-7334 for contact inf
19660124-PDF-138        703-632-7356     be directed to 703-632-7356 or 703-632-7334
19660124-PDF-154        703-632-7334    703-632-7356 or 703-632-7334. Follow the evi
19660124-PDF-633        703-632-8194     be directed to 703-632-8194.DNA Examination
19664312-PDF-671        703-632-8446     be directed to 703-632-8446. Questions conc
19664312-PDF-754        703-632-7572     be directed to 703-632-7572. Follow the evi
19670442-PDF-280        703-632-8446    to the DNAUI at 703-632-8446. As necessary:D
19672393-PDF-408        703-632-7572    o the DNAUII at 703-632-7572.State and local
19672393-PDF-579        703-632-7572    encies may call 703-632-7572 to discuss the
19674361-PDF-330        703-632-7582    gram Manager at 703-632-7582 for questions r
19700243-PDF-145        703-632-7572    e Laboratory at 703-632-7572 prior to submit
19705414-PDF-741        703-985-2400    be directed to  703-985-2400. Questions conc
19705414-PDF-834        703-985-1388     be directed to 703-985-1388. FBI entities m
19709328-PDF-391        703-632-7626    e Laboratory at 703-632-7626 each time an ex
19709328-PDF-659        703-632-7626     be directed to 703-632-7626. Explosives are
19711159-PDF-802        703-632-7626     be directed to 703-632-7626. Follow the evi
19715179-PDF-429        202-323-3300    tions Center at 202-323-3300. Back to the to
19715179-PDF-697        703-632-8449     be directed to 703-632-8449. Follow the evi
19723239-PDF-186        703-632-8442     be directed to 703-632-8442. Follow the evi
```

Conclusion

In this report, I managed to extract and analyze the given file and given an Autopsy to verify and recover multiple files. I checked the md5 and sha of the file, the metadata, The conclusion was that the given file was very healthy and did not have malware or corrupted files. There were a lot of Png and jpeg files. Only about 2 or 3 pdg files and a lot of random images. The carving and recovery tools helped me in examining the data such as searches, instagram logs and telephone numbers. In conclusion, this file does not have a significant value in the investigation.

References

Chain of custody and Critical Infrastructure Systems. (n.d.). https://www.cisa.gov/sites/default/files/publications/cisa-insights_chain-of-custody-and-ci-systems_508.pdf

Magicrescue: Kali linux tools. Kali Linux. (2023a, March 8). https://www.kali.org/tools/magicrescue/

Scalpel: Kali linux tools. Kali Linux. (2023b, March 8). https://www.kali.org/tools/scalpel/

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